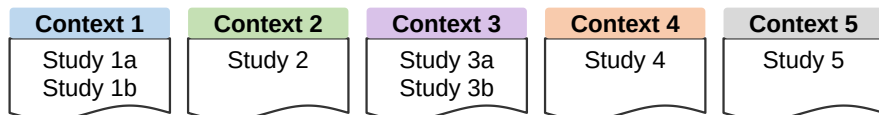


Individual-level data within studies

Microbiome profiles and metadata



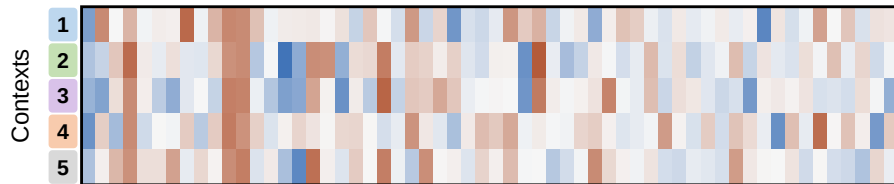
PALM association analysis

Study-level association summaries



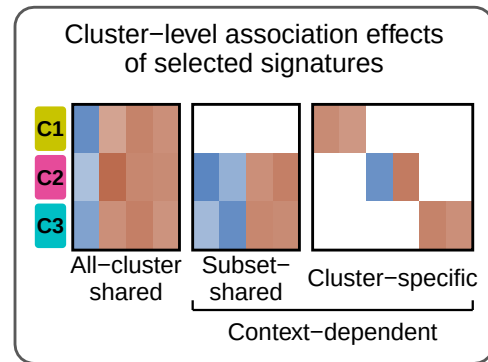
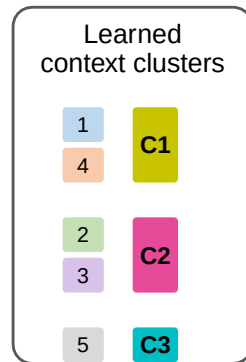
Summary aggregation

Context-level association summaries ($\hat{\beta}$, SE)



Candidate microbial features

SMESH outputs



Precision-weighted structure-learning meta-analysis

Stage 1: learn context clusters

Stage 2: refine signature-sharing patterns